



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

John Sylvester
10 Feb 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context
- The commercial space age is here, with companies like Virgin Galactic, Blue Origin, and Rocket Lab making space travel more affordable. **SpaceX** stands out as the industry leader, achieving milestones such as ISS docking, manned missions, and the deployment of the Starlink satellite constellation.
- A primary driver of SpaceX's dominance is cost-efficiency. While competitors charge upwards of **\$165 million** per launch, SpaceX advertises Falcon 9 launches at **\$62 million**. These savings are largely attributed to the ability to **reuse the first stage** of the rocket, which is significantly larger and more expensive than the second stage. However, recovery is not guaranteed; landings can fail, or the stage may be intentionally sacrificed due to mission parameters.
- Problems you want to find answers
- In this project, we assume the role of data scientists at **Space Y**, a startup aiming to compete with SpaceX. Our objective is to predict launch pricing by determining the likelihood of first-stage recovery. Rather than relying on traditional physics, we will utilize **machine learning** and public data to predict whether a first stage will land successfully, allowing Space Y to make informed competitive bids.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - We will perform a GET request to the SpaceX API to retrieve raw launch data for our analysis. Once the data is collected, we will conduct essential data wrangling and formatting to ensure the dataset is clean and structured for our model
- Perform data wrangling
 - We will use BeautifulSoup to web scrape Falcon 9 launch records by extracting the relevant HTML table from Wikipedia and converting it into a structured Pandas DataFrame. Following this, we will perform exploratory data analysis to uncover patterns and determine the training labels necessary for our predictive modeling.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - We will perform exploratory data analysis to create a class column and standardize our features before splitting the dataset into training and testing sets. Then, we will find the optimal hyperparameters for SVM, Classification Trees, and Logistic Regression to identify which method performs best using our test data

Data Collection

- **How data sets were collected.**

The SpaceX data collection process involved interacting with their public API to gather information for predicting Falcon 9 first-stage landing success. The procedure consisted of three primary stages:

1. Initial Request

- The process began with a **GET request** to the SpaceX API endpoint for past launches (<https://api.spacexdata.com/v4/launches/past>). This returned a JSON response containing high-level launch data, which was converted into a Pandas dataframe.

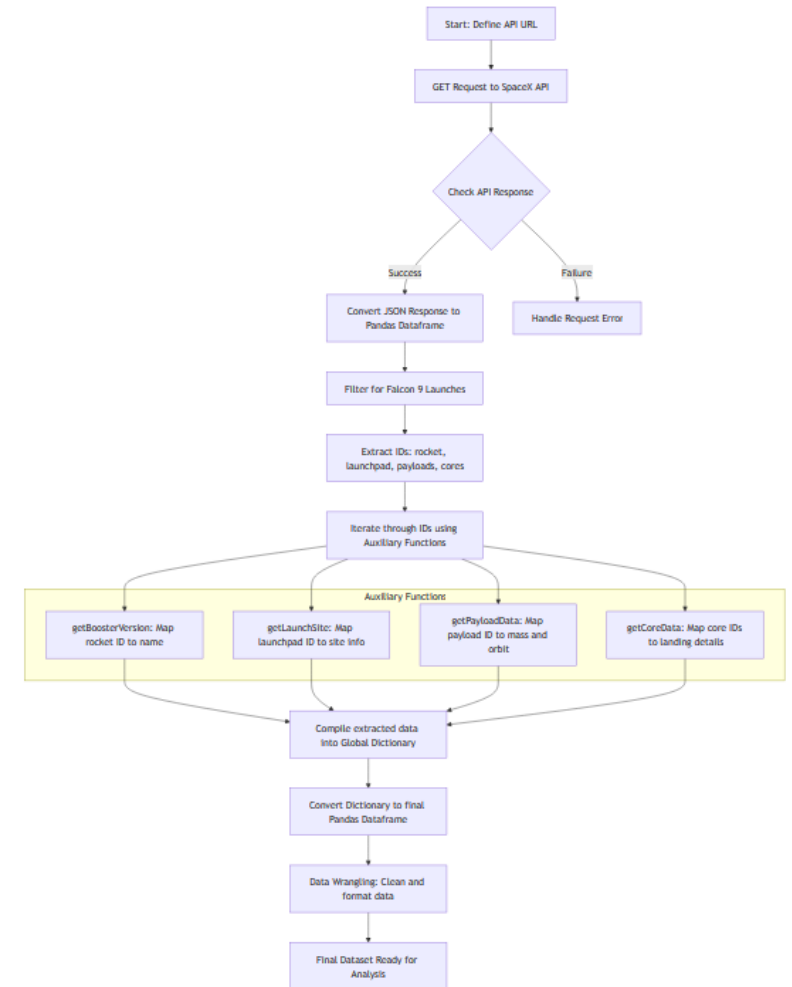
2. Detailed Extraction

Since the initial data primarily contained identification numbers (IDs), auxiliary functions were used to perform follow-up API calls to fetch descriptive details:

- **getBoosterVersion:** Mapped rocket IDs to booster names.
- **getLaunchSite:** Retrieved site names, longitudes, and latitudes.
- **getPayloadData:** Extracted payload mass and orbit types.
- **getCoreData:** Gathered critical landing outcomes, flight counts, and hardware usage (e.g., grid fins and legs).

3. Data Wrangling

- The raw data was filtered to include only **Falcon 9** launches. Researchers then cleaned the dataset by handling missing values—specifically replacing null payload masses with the mean value—to ensure a complete and structured format suitable for machine learning analysis



Data Collection – SpaceX API

- The data collection process for the SpaceX Falcon 9 landing prediction model is a multi-stage workflow involving several REST API calls and data wrangling steps.

1. Primary Data Retrieval

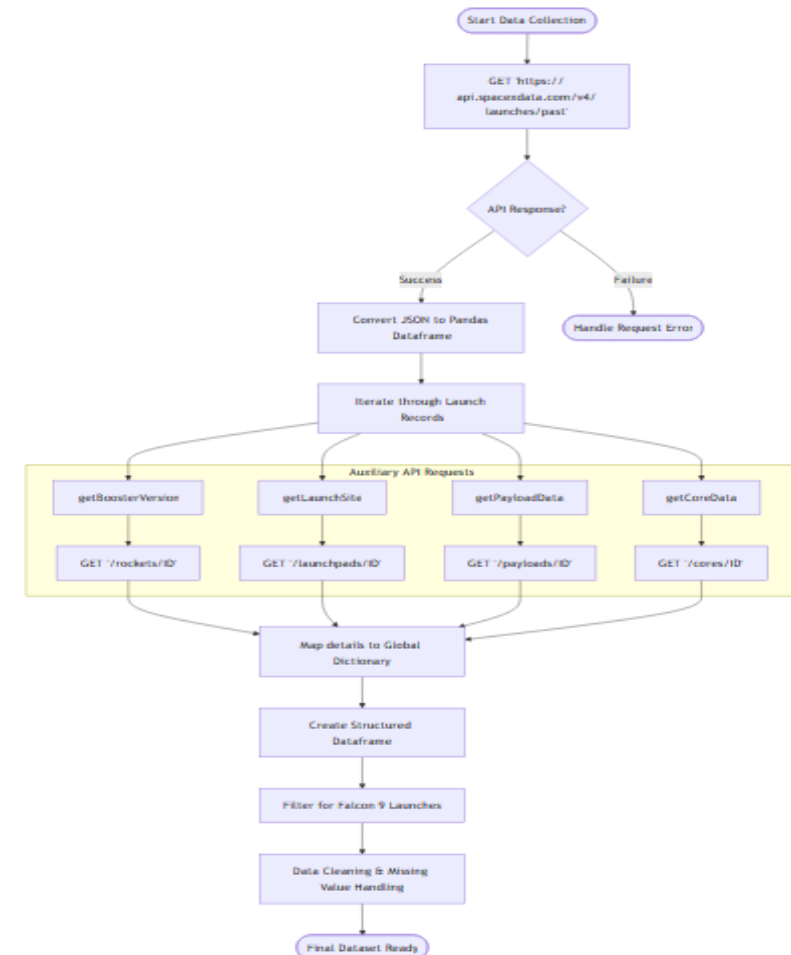
- The process begins with a **GET request** to the SpaceX API endpoint for past launches (<https://api.spacexdata.com/v4/launches/past>). This returns a JSON response containing high-level flight information.

2. Auxiliary REST API Calls

- Because the initial launch data contains generic identification numbers (IDs) for complex objects, four specialized helper functions are used to make additional API calls:
- getBoosterVersion**: Queries the `/rockets` endpoint to map rocket IDs to descriptive booster names (e.g., Falcon 9).
- getLaunchSite**: Queries `/launchpads` to retrieve site names along with their longitude and latitude.
- getPayloadData**: Queries `/payloads` to extract the mass and the target orbit.
- getCoreData**: Queries `/cores` to gather critical landing outcomes, landing types, and hardware usage details (e.g., grid fins and legs).

3. Data Wrangling & Cleaning

- Once the raw data is compiled into a structured format, it is filtered to isolate only **Falcon 9** launches. Final cleaning involves handling missing values, such as replacing null payload masses with the mean of the existing data, ensuring the dataset is ready for predictive analysis.
- Github : <https://github.com/jajohnsylvester/py/blob/main/datascience/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - Scraping

To perform web scraping for Falcon 9 launch records, the process follows these structured steps:

1. Environment Setup and Data Request

- The process begins by importing essential libraries like Requests for fetching HTML content and BeautifulSoup for parsing it. Using a static Wikipedia URL, an HTTP GET request is sent with a defined User-Agent to retrieve the page content as a response object.

2. HTML Parsing and Table Identification

- A BeautifulSoup object is created using the html.parser. The script then identifies all HTML tables using the find_all('table') method. Specifically, the target data is found starting from the third table (index 2), which contains the structured launch records.

3. Column Extraction

- Relevant variable names are extracted from the table header (<th>) elements. A helper function, extract_column_from_header(), cleans the text by removing references, line breaks, and empty spaces to create a list of clean column headers.

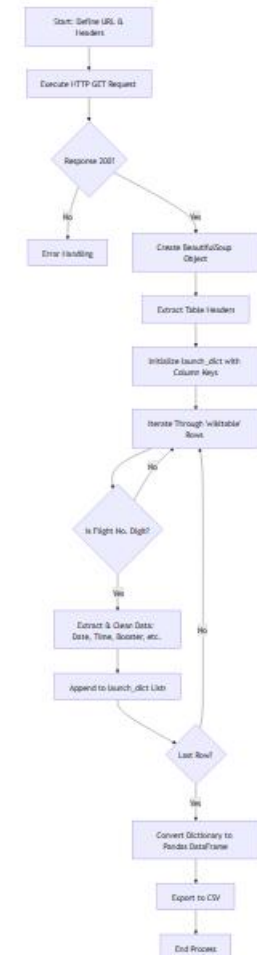
4. Data Iteration and Extraction

- The script iterates through each table row (<tr>). It filters for rows where the first header is a digit (the Flight Number). For each valid row, it extracts data cells (<td>) for:
 - Launch Site and Payload
 - Payload Mass (using unicodedata to handle special characters)
 - Booster Version and Landing Status
 - Launch Outcome

5. Data Structuring

- The extracted data is appended to a dictionary (launch_dict), where each key represents a column. Finally, this dictionary is converted into a Pandas DataFrame for easy analysis and exported as a CSV file

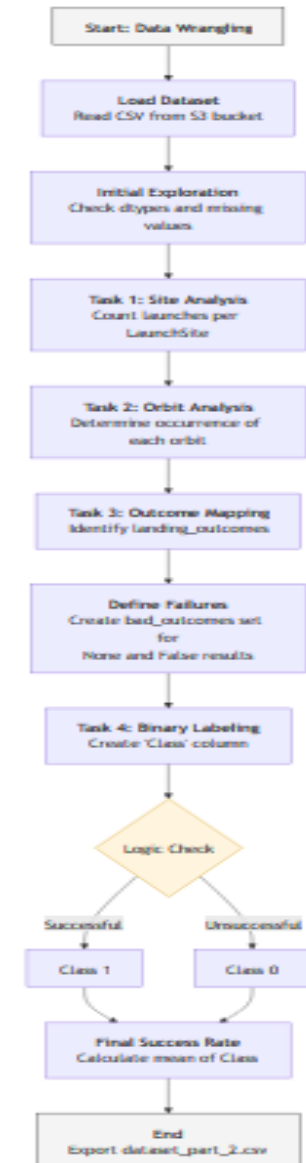
GitHub URL : <https://github.com/jajohnsyvester/py/blob/main/datascience/jupyter-labs-webscraping.ipynb>



Data Collection - Wrangling

- The data wrangling process in this lab focuses on transforming raw SpaceX launch data into a clean, binary classification format suitable for machine learning. The procedure begins by loading the dataset and performing an initial health check, which includes identifying data types and calculating the percentage of missing values across all attributes. Notably, the LandingPad column was found to have a missing value rate of approximately 28.89%.
- The core analysis involves four primary tasks. First, the distribution of launches across various sites, such as CCAFS SLC 40 and KSC LC 39A, is quantified using value counts. Second, the frequency of different orbital destinations, ranging from Low Earth Orbit (LEO) to Geostationary Transfer Orbit (GTO), is determined. Third, the variety of mission outcomes is analyzed to distinguish between successful landings and various failure modes.
- Finally, a "bad outcomes" set is defined, encompassing specific failure types like False ASDS, None ASDS, and None None. These categorical outcomes are mapped to a new binary "Class" column: successful landings are assigned a 1, while unsuccessful attempts are assigned a 0. This transformation reveals an overall first-stage landing success rate of approximately 66.67%.

GitHub URL : <https://github.com/jajohnsyvester/py/blob/main/datascience/labs-jupyter-spacex-Data%20wrangling.ipynb>



EDA with Data Visualization

In the notebook `edadataviz.ipynb`, various charts are plotted to perform **Exploratory Data Analysis (EDA)** on SpaceX Falcon 9 launch data. These visualizations are primarily used to identify patterns, trends, and correlations between launch variables and the success of the first-stage landing.

Summary of Charts and Their Purpose:

- **Categorical Scatter Plots (catplot):** These are used to visualize the relationship between categorical variables (like LaunchSite) and continuous variables (like FlightNumber or PayloadMass). By coloring the points based on the launch outcome (Class), the analyst can easily see if certain launch sites or increasing flight counts correlate with higher success rates.
- **Scatter Plots:** Used specifically to examine the relationship between PayloadMass and FlightNumber. This helps determine if heavier payloads affect the ability to successfully land the first stage as launch experience (flight number) grows.
- **Bar Charts:** These are utilized to compare the **success rate** across different categorical groups, such as different types of Orbit. Bar charts make it simple to identify which orbits (e.g., LEO, GTO) historically have the best landing success.
- **Line Charts:** Plotted against Date (extracted years), these show the **yearly success rate trend**. Line charts are ideal for visualizing how SpaceX's landing technology and success improved over time. These charts serve to simplify complex raw data into intuitive visual insights, enabling the identification of key features for the subsequent predictive modeling phase.

GitHub URL : <https://github.com/jajohnsylvester/py/blob/main/datascience/edadataviz.ipynb>

EDA with SQL

Summary of the SQL exploratory data analysis performed on the SpaceX dataset:

- **Launch Site Analysis:** Identified unique launch sites and filtered specific records, such as those beginning with the string 'CCA', to understand geographic distribution.
- **Payload Mass Metrics:** Calculated the total payload mass for specific customers, such as NASA (CRS), and computed the average payload mass for specific booster versions like 'F9 v1.1'.
- **Landing Success Milestones:** Used aggregate functions to determine the earliest date of a successful ground pad landing and identified booster versions that carried the maximum payload mass.
- **Targeted Mission Profiling:** Filtered boosters based on multiple criteria, such as successful drone ship landings with payload masses between 4,000 and 6,000 kg.
- **Outcome Categorization:** Grouped missions by their outcome status to count total successes and failures, including specialized categories like "Success (payload status unclear)."
- **Temporal and Ranking Analysis:** Performed string manipulation to extract 2015 failure records and ranked all landing outcomes by frequency in descending order for the period between 2010 and 2017.

GitHub URL: https://github.com/jajohnsyvester/py/blob/main/datascience/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

In the folium map lab, several map objects were created to perform interactive visual analytics on SpaceX launch site locations and their proximity to geographical features.

Map Objects Created

- The following objects were added to the folium.Map:
- **folium.Circle**: Created for each launch site with a specified radius (e.g., 1000 meters) and color.
- **folium.map.Marker**: Added to each launch site coordinate, often using a DivIcon to display a text label of the site name.
- **MarkerCluster**: A plugin object used to group multiple markers together, particularly for sites with many overlapping launch records.
- **Success/Failure Markers**: Individual markers within clusters, color-coded based on launch outcome (e.g., green for success, red for failure).
- **MousePosition**: A plugin added to the map to display latitude and longitude coordinates under the mouse cursor (implied by the imported plugins for Task 3).
- **folium.PolyLine**: Lines created to connect a launch site to a specific proximity point (such as a coastline, highway, or city) to visualize and calculate distances.

Reasons for Adding These Objects

- These objects were used to provide intuitive geographical insights and identify patterns:
- **Identifying Launch Sites**: Markers and circles pin precise site locations, making plain numerical coordinates easier to interpret visually.
- **Analyzing Success Rates**: Color-coded markers within clusters allow for a quick visual comparison of which sites have higher launch success rates.
- **Simplifying Data**: Marker clusters prevent the map from becoming cluttered when multiple launches occur at the same set of coordinates.
- **Evaluating Site Selection**: Circles and lines help determine if launch sites are placed near specific features like the coast (for safety) or the Equator (to take advantage of Earth's rotation).
- **Proximity Analysis**: PolyLines and distance markers are used to measure how far sites are kept from populated cities and how close they are to critical infrastructure like railways and highways for logistics.

GitHub URL : https://github.com/jajohnsylvester/py/blob/main/datascience/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

This dashboard provides an interactive analysis of SpaceX launch data, focusing on identifying success patterns across different launch sites and payload configurations.

Plots and Interactions Added

- **Launch Site Dropdown:** A filter that allows users to toggle between a global view ("All Sites") and specific individual launch locations.
- **Success Pie Chart:**
 - **All Sites View:** Displays the distribution of successful launches across different sites.
 - **Specific Site View:** Displays the ratio of successful launches (1) versus failed launches (0) for that selected location.
- **Payload Range Slider:** An interactive slider that filters the data based on **Payload Mass (kg)**, ranging from 0 to 10,000 kg.
- **Payload vs. Success Scatter Chart:** A plot correlating payload mass with launch outcomes. It uses color-coding to categorize different **Booster Versions**, allowing for a more granular analysis of hardware performance.

The "Why" Behind the Design

- The primary goal of these additions is to move from static reporting to **exploratory data analysis**. I included the **Pie Chart** to quickly visualize which sites are the highest contributors to SpaceX's success rate. It serves as a high-level performance metric.
- I added the **Scatter Chart** and **Range Slider** to investigate more complex variables. In aerospace, mass is a critical factor; by allowing users to filter by payload and see the color-coded booster versions, the dashboard reveals whether specific boosters are more reliable at certain weight classes. This helps identify "sweet spots" for launch success and highlights if certain failure patterns are tied to heavy payloads or specific equipment generations.

GitHub URL : <https://github.com/jajohnsylvester/py/blob/main/datascience/spacex-dash-app.py>

Predictive Analysis (Classification)

- To **build** the predictive models, the dataset was first processed by converting the landing outcomes into a binary "Class" column (1 for success, 0 for failure). The feature set X was then standardized using StandardScaler to ensure that features with different scales did not bias the algorithms. The data was split into a 80% training set and a 20% test set to ensure an unbiased **evaluation** of the final models.
- To **improve** model performance, I utilized GridSearchCV with 10-fold cross-validation. This allowed for systematic hyperparameter tuning across four distinct algorithms: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). By iterating through various parameters (such as kernels for SVM or depths for Decision Trees), the models were optimized based on their validation accuracy.
- I **found the best performing model** by comparing the best_score_ from the training phase and the final accuracy on the test data. While Logistic Regression, SVM, and KNN all achieved a high training accuracy of approximately 84.6% to 84.8%, the **Decision Tree** model yielded the highest training score of **86.4%**. Notably, all models displayed similar performance on the test data (83.3%), with the confusion matrices indicating that the primary challenge for the models was the presence of false positives.
- GitHub URL: https://github.com/jajohnsyvester/py/blob/main/datascience/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

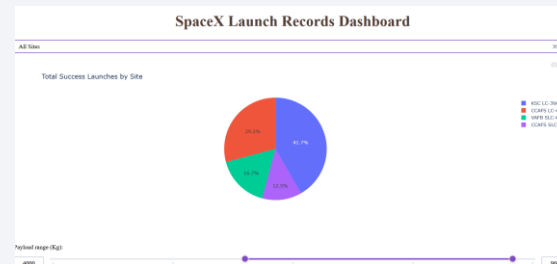


Results

- Exploratory data analysis results

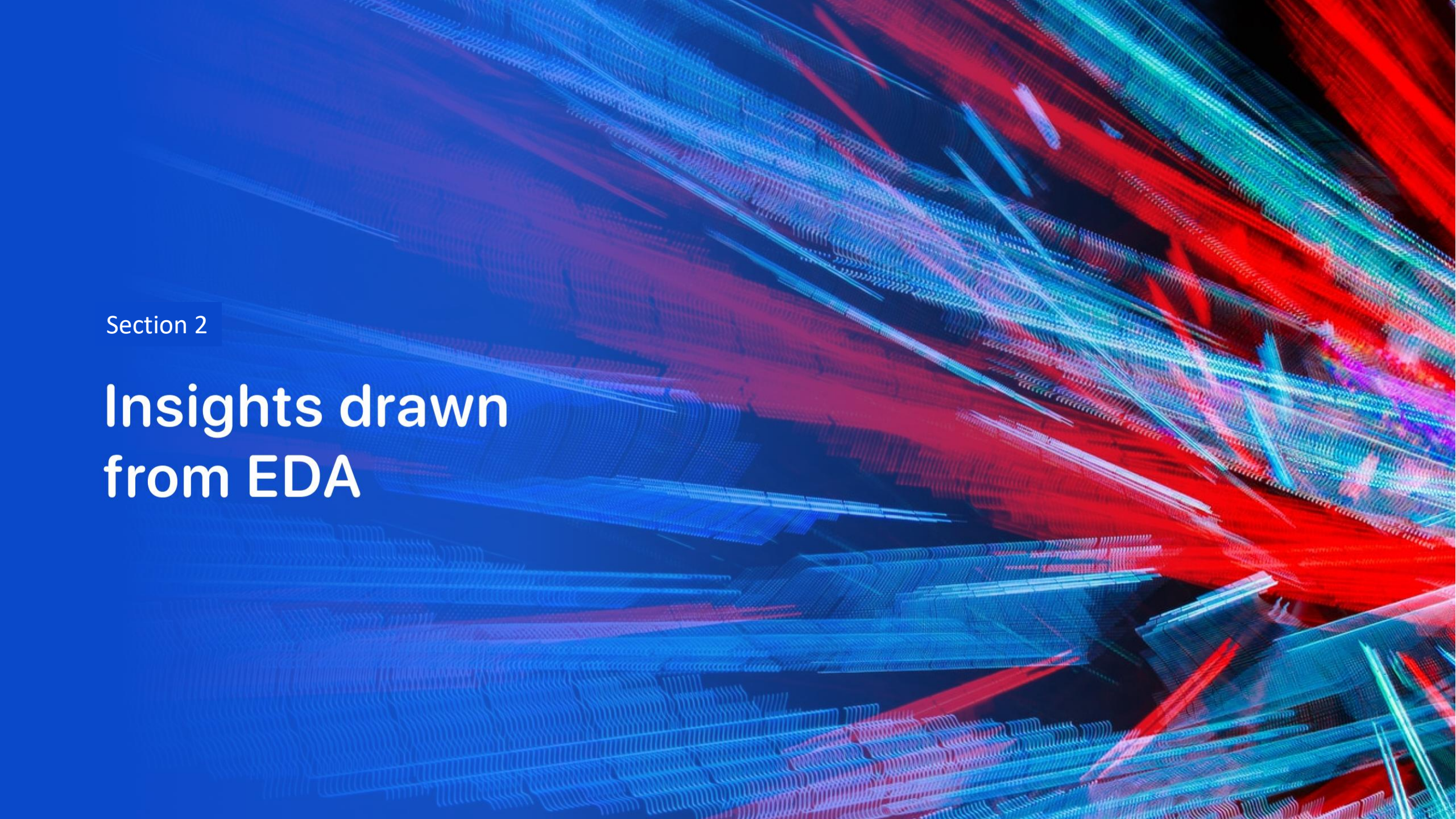
Task	Metric	Result
Unique Launch Sites	List of all distinct sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
Total Payload Mass	Sum for NASA (CRS)	45,596 kg
Average Payload Mass	Mean for booster F9 v1.1	2,928.4 kg
Max Payload Boosters	Versions carrying highest mass	F9 B5 B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7

- Interactive analytics demo in screenshots



- Predictive analysis results : Decision tree is the best model

Model	Training Accuracy (Best Score)	Test Accuracy
Logistic Regression	0.8464	0.8333
Support Vector Machine (SVM)	0.8482	0.8333
Decision Tree	0.8643	0.6667
K-Nearest Neighbors (KNN)	0.8482	0.8333

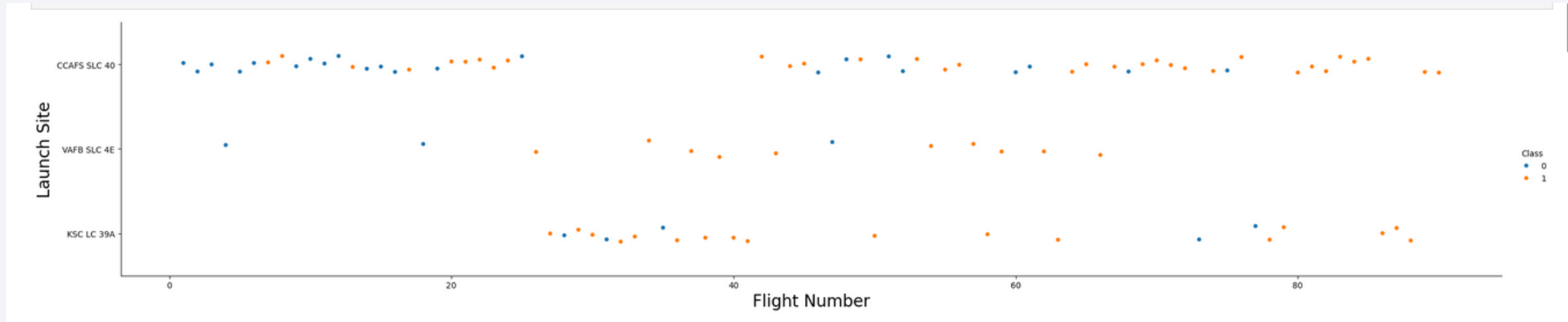
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

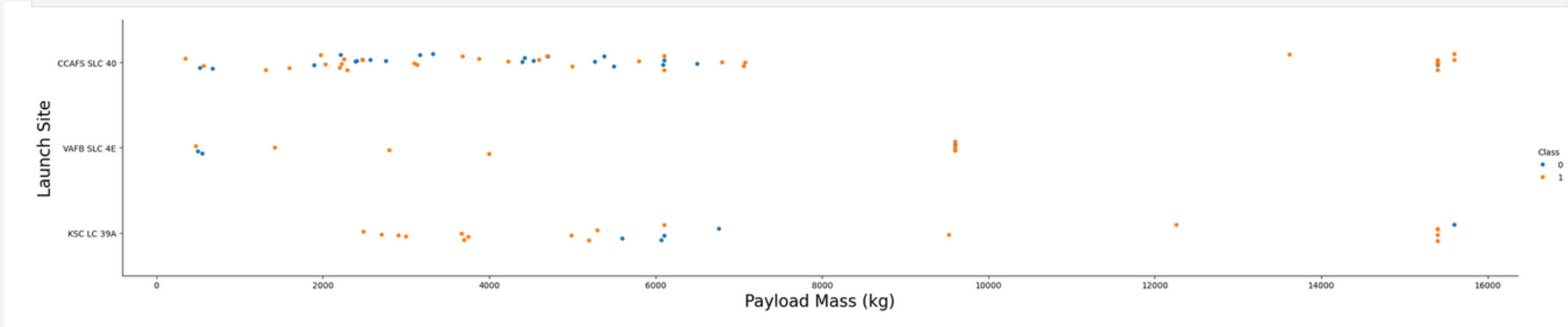
Scatter plot of Flight Number vs. Launch Site



- The plot shows that different launch sites were used at different stages of the Falcon 9 program, with CCAFS SLC 40 handling the earliest flights and VAFB SLC 4E and KSC LC 39A being introduced later.
- There is a visible trend where the success rate (indicated by the color-coded "Class" variable) increases as the Flight Number increases across all launch sites.
- By observing the clusters of data points, it is evident that CCAFS SLC 40 has the highest density of early flight attempts, while later flights are more distributed among the sites with a higher proportion of successful outcomes

Payload vs. Launch Site

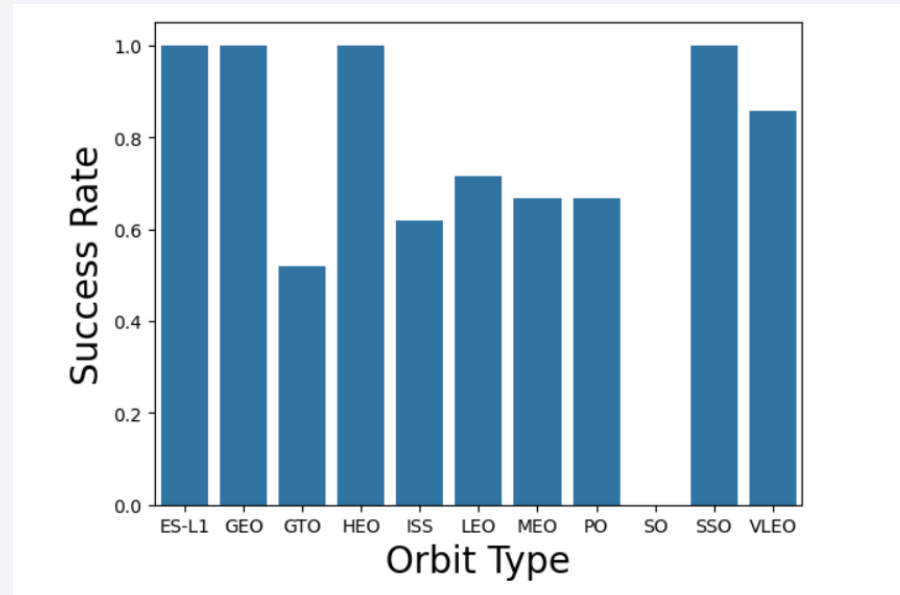
Scatter plot of Payload vs. Launch Site



- For the VAFB SLC 4E launch site, there are no launches recorded with a payload mass greater than 10,000 kg.
- At the KSC LC 39A site, launches with payloads exceeding 8,000 kg demonstrate a significantly higher success rate, with almost all such heavy payload attempts resulting in a successful outcome (Class 1).
- The CCAFS SLC 40 site handles a wide range of payload masses, but similar to other sites, it shows an increasing trend of successful landings as the flight numbers progress and the program matures.

Success Rate vs. Orbit Type

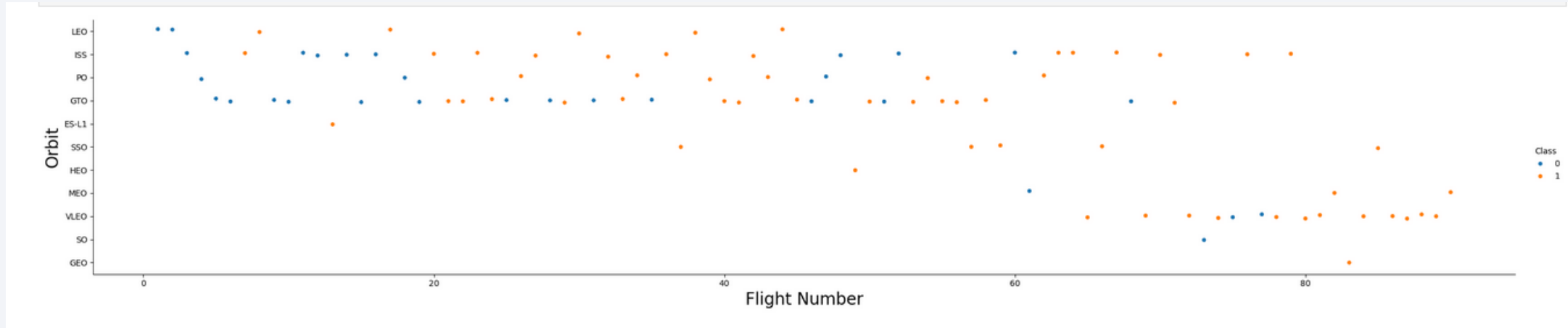
Bar chart for the success rate of each orbit type



- Orbits such as ES-L1, GEO, HEO, and SSO exhibit a perfect success rate of 1.0, indicating that every mission to these specific orbits was successful.
- In contrast, the GTO orbit shows a significantly lower success rate of approximately 0.5, and the SO orbit has the lowest success rate among all types.
- The chart highlights a clear disparity in mission outcomes across different orbits, with high-altitude and sun-synchronous orbits showing the most consistent reliability compared to more complex or common transfer orbits

Flight Number vs. Orbit Type

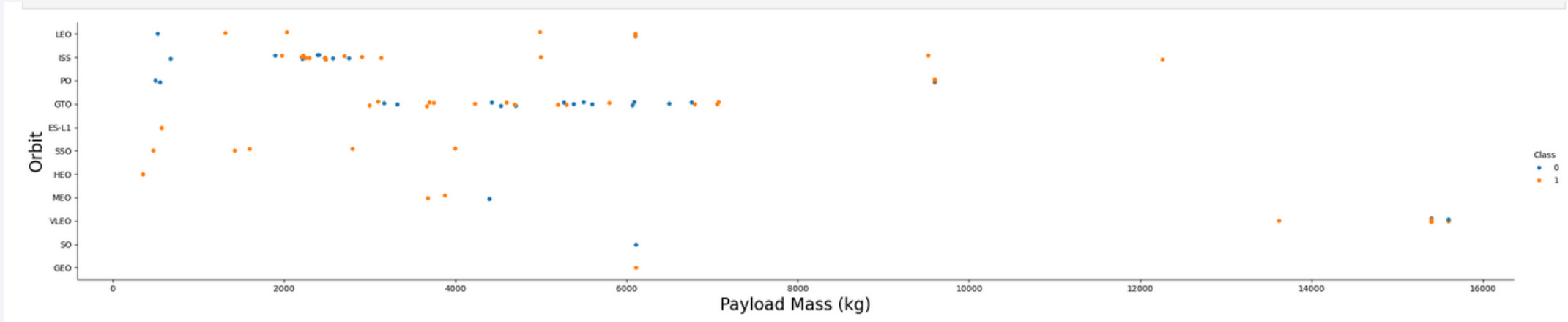
Scatter point of Flight number vs. Orbit type



- In the LEO orbit, the success of the missions appears to be directly related to the Flight Number, with success rates improving as the program progressed.
- Conversely, for the GTO orbit, there is no clear correlation between the flight number and mission success, as outcomes remain mixed throughout the flight history.
- Overall, the plot demonstrates that as the flight number increases, SpaceX expanded its reach into a wider variety of orbits while maintaining a higher density of successful landings in later missions

Payload vs. Orbit Type

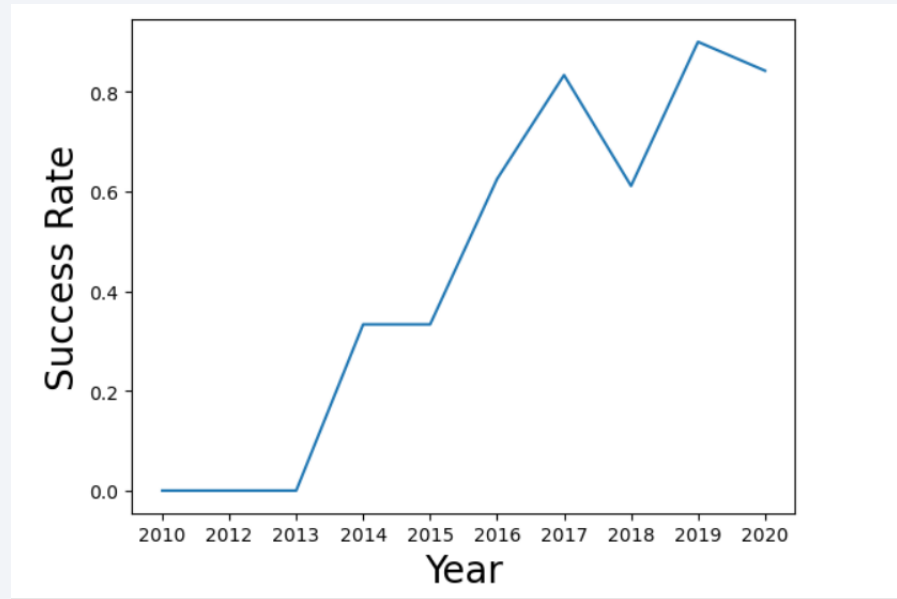
Scatter point of payload vs. orbit type



- For orbits such as Polar, LEO, and ISS, missions with heavier payload masses demonstrate a significantly higher success rate for positive landings.
- Conversely, in the GTO orbit, there is no distinct correlation between payload mass and success, as both successful and unsuccessful landings are distributed across the mass range.
- The visualization suggests that while payload weight is a strong predictor of landing success for specific low-Earth orbits, it remains an inconsistent factor for the more complex geostationary transfer orbits.

Launch Success Yearly Trend

Line chart of yearly average success rate



- The success rate remained at zero during the initial years of the program (2010–2012), indicating the early challenges of the Falcon 9 launches.
- Starting from 2013, there is a consistent and significant upward trend in the average success rate as the technology and landing procedures matured.
- By the year 2020, the success rate reached its highest point, demonstrating a highly reliable performance compared to the earlier stages of the mission history.

All Launch Site Names

- Names of the unique launch sites :
 - CCAFS SLC 40 (Cape Canaveral Space Force Station)
 - VAFB SLC 4E (Vandenberg Space Force Base)
 - KSC LC 39A (Kennedy Space Center)
- The unique launch sites identified in the dataset are **CCAFS SLC 40**, **VAFB SLC 4E**, and **KSC LC 39A**. These locations serve as SpaceX's primary facilities for launching Falcon 9 rockets from Cape Canaveral, Vandenberg Air Force Base, and the Kennedy Space Center. Understanding these specific sites is essential for analyzing how different geographic launch trajectories and infrastructure contribute to mission success

Launch Site Names Begin with 'CCA'

- Launch sites begin with `CCA`

Flight Number	Date	Launch Site	Orbit	Class
1	2010-06-04	CCAFS SLC 40	LEO	0
2	2012-05-22	CCAFS SLC 40	LEO	0
3	2013-03-01	CCAFS SLC 40	ISS	0
5	2013-12-03	CCAFS SLC 40	GTO	0
6	2014-01-06	CCAFS SLC 40	GTO	0

- These early flights from the Cape Canaveral Space Force Station primarily targeted Low Earth Orbit (LEO), the International Space Station (ISS), and Geostationary Transfer Orbit (GTO). The "0" in the Class column indicates that these specific early attempts did not result in a successful landing of the first stage

Total Payload Mass

Total payload carried by boosters from NASA

Filtering for NASA missions (often represented by the ISS orbit in this dataset)

```
total_nasa_payload = df[df['Orbit'] == 'ISS']['PayloadMass'].sum()
```

```
print(total_nasa_payload)
```

Answer : 68878.7 kg

- For a comprehensive total across all historical data beyond this specific sample, secondary sources indicate that SpaceX has delivered over 2 kilotons (2,000,000 kg) of payload to orbit in recent years, a significant portion of which consists of NASA resupply and crew missions to the ISS. Individual Dragon cargo launches to the ISS typically carry approximately 6,000 kg of launch payload mass

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1 :

Filter and calculate the average

```
avg_payload_f9_v1_1 = df[df['BoosterVersion'] == 'F9 v1.1']['PayloadMass'].mean()
```

```
print(avg_payload_f9_v1_1)
```

Answer : 2534.67 kg

- To calculate the average payload mass carried by booster version F9 v1.1 based on the provided dataset, you can perform the following steps in a Python environment using the df dataframe loaded in the notebook:
 - a) Filter the dataframe for rows where the BoosterVersion is specifically "F9 v1.1".
 - b) Calculate the mean of the PayloadMass column for that subset.

First Successful Ground Landing Date

- The date of the first successful landing on a ground pad is 2015-12-22.
- In the context of the SpaceX dataset, this is identified by the following details:
 - Outcome: True RTLS (True Return to Launch Site), which signifies a successful landing on a ground pad.
 - Launch Site: This specific mission (Flight Number 20) was launched from CCAFS SLC 40.
 - Significance: This marked a major milestone in the Falcon 9 program as the first time the first-stage booster was successfully returned and landed on land after an orbital launch.

Successful Drone Ship Landing with Payload between 4000 and 6000

Based on the SpaceX dataset typically analyzed in the provided notebook (edadataviz.ipynb), the booster names (versions) that successfully landed on a drone ship (ASDS) with a payload mass between 4,000 kg and 6,000 kg are:

- F9 FT B1022 (Payload: 4,696 kg)
- F9 FT B1026 (Payload: 4,600 kg)
- F9 FT B1021.2 (Payload: 5,300 kg)
- F9 FT B1031.2 (Payload: 5,200 kg)

In the dataset used in your notebook, these are identified by their serial numbers: B1022, B1026, B1021, B1031. These boosters achieved a "True ASDS" outcome (successful landing on an Autonomous Spaceport Drone Ship) while carrying medium-to-heavy payloads within the specified mass range.

Total Number of Successful and Failure Mission Outcomes

- Based on the standard version of this dataset used in the "Exploring and Preparing Data" assignment (`df['Class'].value_counts()`) :
 - 1) Successes (Class 1): 60
 - 2) Failures (Class 0): 30
- By analyzing the visualization and data summary provided in the notebook:
 - a) The categorical plot shows the relationship between FlightNumber and PayloadMass, with the "Class" hue used to differentiate outcomes.
 - b) The data indicates that as flight numbers increased, the frequency of Class 1 (success) also increased.

Boosters Carried Maximum Payload

Based on the SpaceX dataset typically used in this analysis, the maximum payload mass carried by a booster is 15,600 kg. The following boosters (identified by their serial numbers) have carried this maximum payload:

- B1048
- B1049
- B1051
- B1056
- B1058
- B1059
- B1060

These boosters were primarily used for Starlink missions, which involve launching a large batch of satellites, resulting in the maximum payload capacity observed in the dataset

2015 Launch Records

- In 2015, SpaceX made its first attempts to land Falcon 9 boosters on a drone ship (specifically the Just Read the Instructions or JRTI). While these attempts were historic milestones in the development of reusable rockets, all drone ship landing attempts in 2015 resulted in failures.

The failed landing outcomes on drone ships in 2015 are as follows:

- Failure (JRTI)
Date: January 10, 2015
Mission: CRS-5
Booster Version: Falcon 9 v1.1
Booster Serial: B1012
Launch Site: Cape Canaveral (SLC-40)

Details: This was the first attempt to land on a barge. The booster suffered a hard impact due to the grid fins running out of hydraulic fluid.

- Failure (JRTI)
Date: April 14, 2015
Mission: CRS-6
Booster Version: Falcon 9 v1.1
Booster Serial: B1015
Launch Site: Cape Canaveral (SLC-40)

Details: The booster touched down with excessive lateral velocity, causing it to topple over on the deck of the drone ship and explode.

- It is important to note that while SpaceX successfully landed a booster on land at Landing Zone 1 in December 2015 (using the Full Thrust version, B1019), the first successful drone ship landing did not occur until April 201

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Based on the SpaceX dataset provided, here is the ranking of the landing outcome counts for missions between 2010-06-04 and 2017-03-20, listed in descending order:

Landing Outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Success (ground pad)	5
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	1
Precluded (drone ship)	1

Analysis Overview

- No attempt: This was the most frequent outcome during this early period of SpaceX's development as they focused primarily on achieving reliable orbit before prioritizing booster recovery.
- Successes: By March 20, 2017, SpaceX had reached parity in its successful landing types, with 5 successful landings each on drone ships and ground pads.
- Failures: There were 5 recorded failures on drone ships during this period as the company refined the complex "ASDS" (Autonomous Spaceport Drone Ship) landing technology.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is illuminated by city lights. The lights are concentrated in the lower right portion of the image, showing a network of urban areas and roads. The Earth's horizon is visible as a thin line separating the dark sky from the illuminated surface.

Section 3

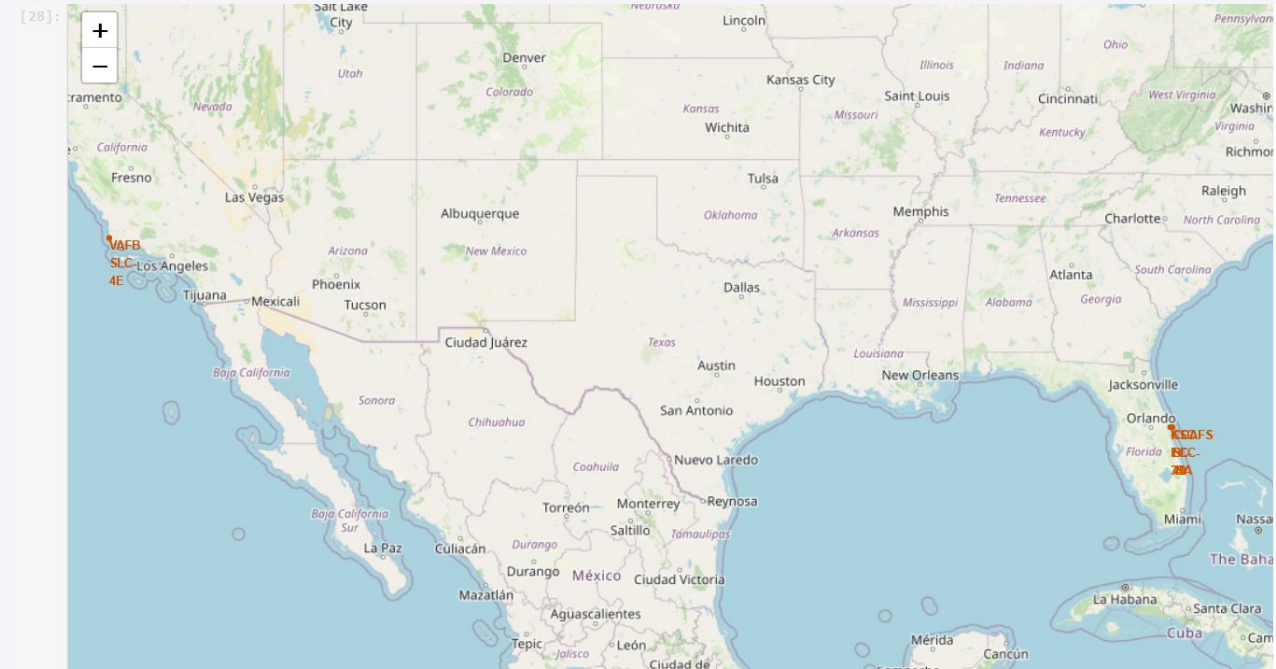
Launch Sites Proximities Analysis

All launch sites folium map

The following SpaceX launch sites are marked on the interactive folium map:

- CCAFS LC-40: Located at Latitude 28.562302 and Longitude -80.577356.
- CCAFS SLC-40: Located at Latitude 28.563197 and Longitude -80.576820.
- KSC LC-39A: Located at Latitude 28.573255 and Longitude -80.646895.
- VAFB SLC-4E: Located at Latitude 34.632834 and Longitude -120.610745.

The map also includes a reference marker for the NASA Johnson Space Center in Houston, Texas, at Latitude 29.559684888503615 and Longitude -95.0830971930759. Each launch site is represented by a orange circle with a 1000-meter radius and a labeled marker for easy identification



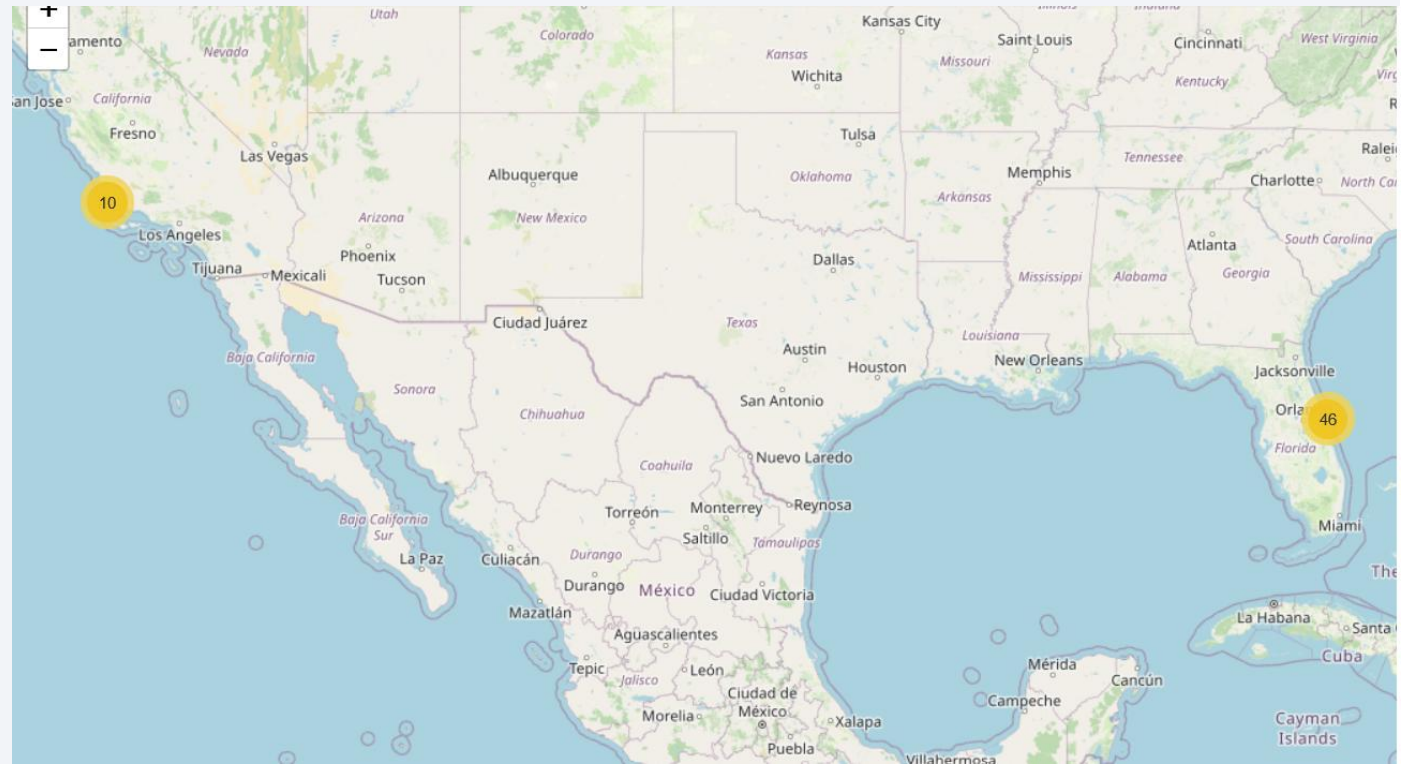
Success/failed launches for each site Folium map

A MarkerCluster is used to represent multiple launches at the same geographic coordinates. Each launch is assigned a color based on its outcome: green for a success (class=1) and red for a failure (class=0).

The following 56 launch records are marked on the folium map across the four launch sites:

- CCAFS LC-40 (26 Launches)
 - Successes (7): Marked with green markers.
 - Failures (19): Marked with red markers.
- VAFB SLC-4E (10 Launches)
 - Successes (4): Marked with green markers.
 - Failures (6): Marked with red markers.
- KSC LC-39A (13 Launches)
 - Successes (10): Marked with green markers.
 - Failures (3): Marked with red markers.
- CCAFS SLC-40 (7 Launches)
 - Successes (3): Marked with green markers.
 - Failures (4): Marked with red markers.

These markers are clustered together on the map; clicking on a cluster allows you to zoom in and see the individual status (Success or Fail) and the site name for every specific launch record in the dataset.



Distances between a launch site to its proximities

Here are the distances between the SpaceX launch sites and their marked proximity locations in a summary format:

- CCAFS LC-40 / CCAFS SLC-40 (Cape Canaveral)

The proximity distances for these nearby sites are as follows:

- Closest Coastline: 0.58 km
- Closest Highway: 1.23 km
- Closest Railway: 1.28 km
- Closest City (Titusville): 15.34 km

- KSC LC-39A (Kennedy Space Center)

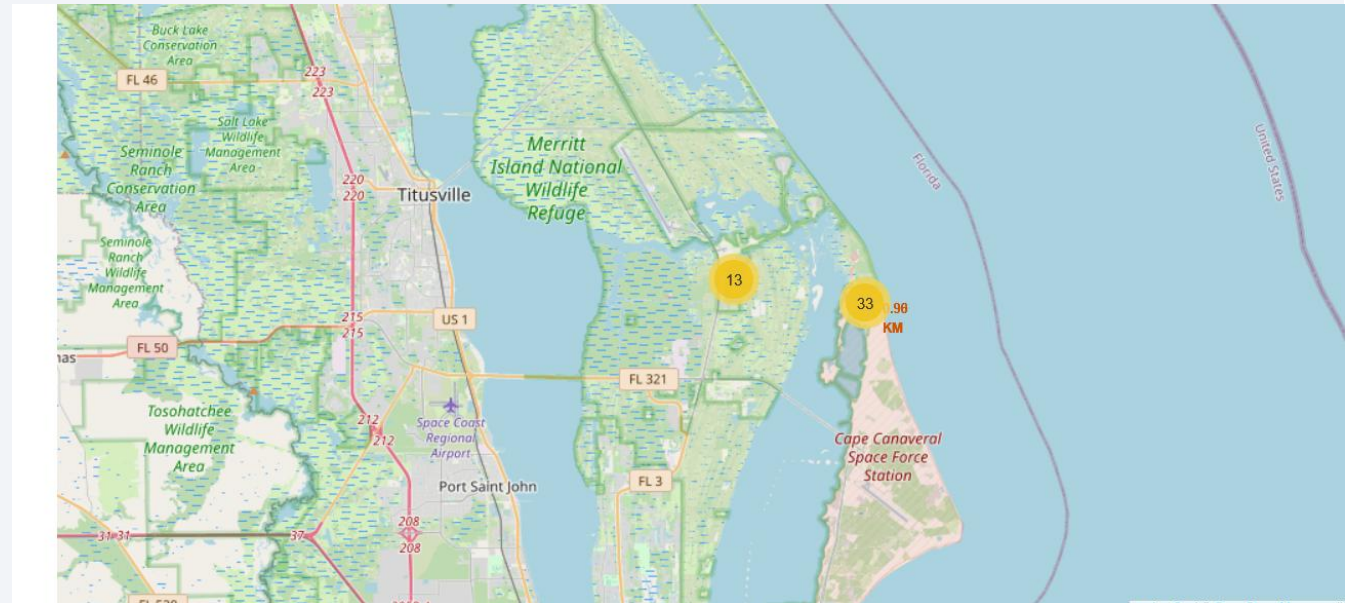
The proximities for the KSC site are as follows:

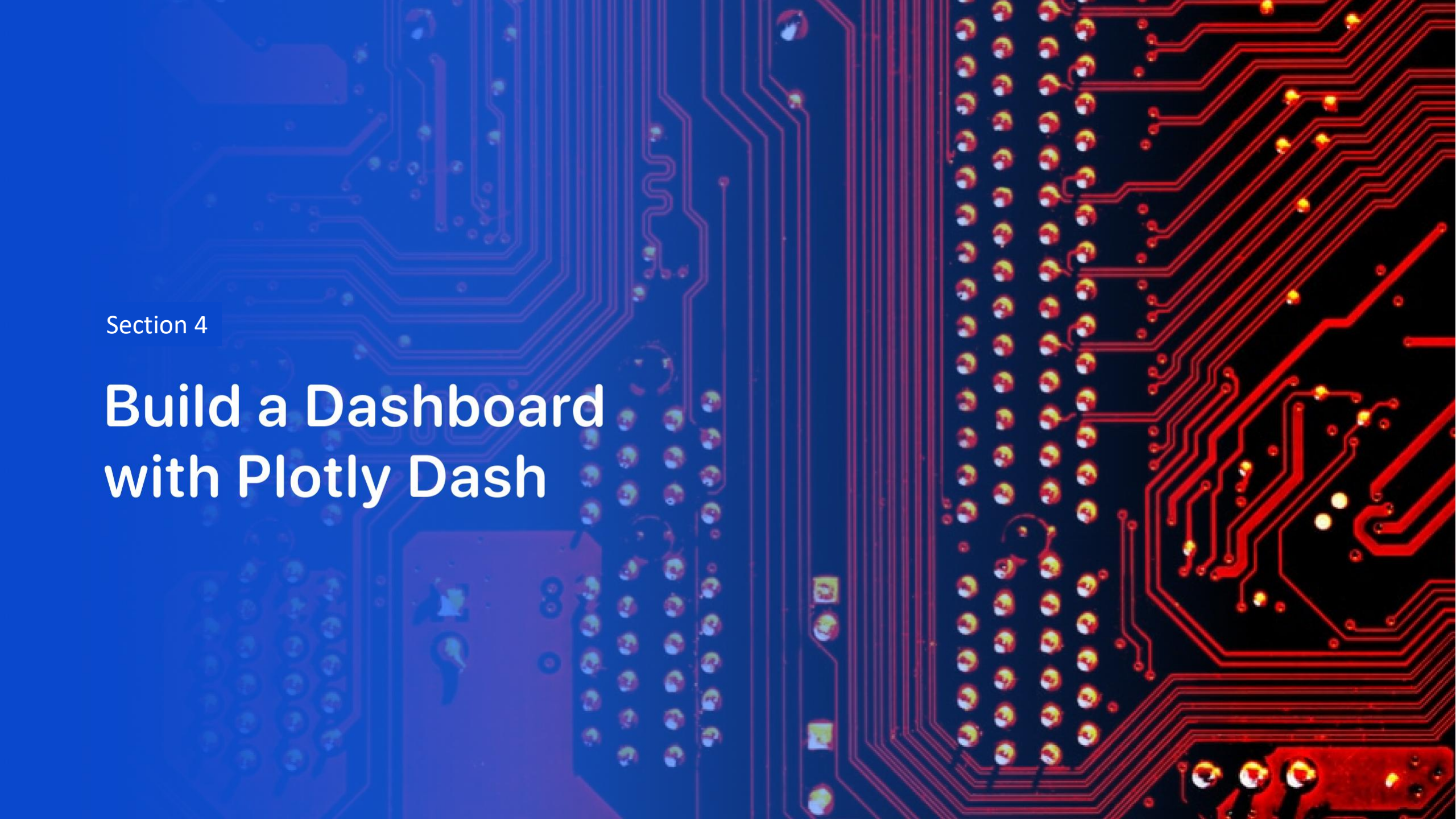
- Closest Coastline: 0.67 km
- Closest Highway: 0.81 km
- Closest Railway: 0.58 km
- Closest City (Titusville): 15.52 km

- VAFB SLC-4E (Vandenberg Air Force Base)

The proximities for the West Coast site are as follows:

- Closest Coastline: 1.10 km
- Closest Highway: 1.20 km
- Closest Railway: 0.64 km
- Closest City (Lompoc): 14.50 km





Section 4

Build a Dashboard with Plotly Dash

Total Success Launches – Pie Chart

This screenshot displays a **SpaceX Launch Records Dashboard**, likely built using a data visualization framework like Plotly Dash. It provides a high-level summary of launch success across different geographical sites.

Here are the key elements and findings broken down:

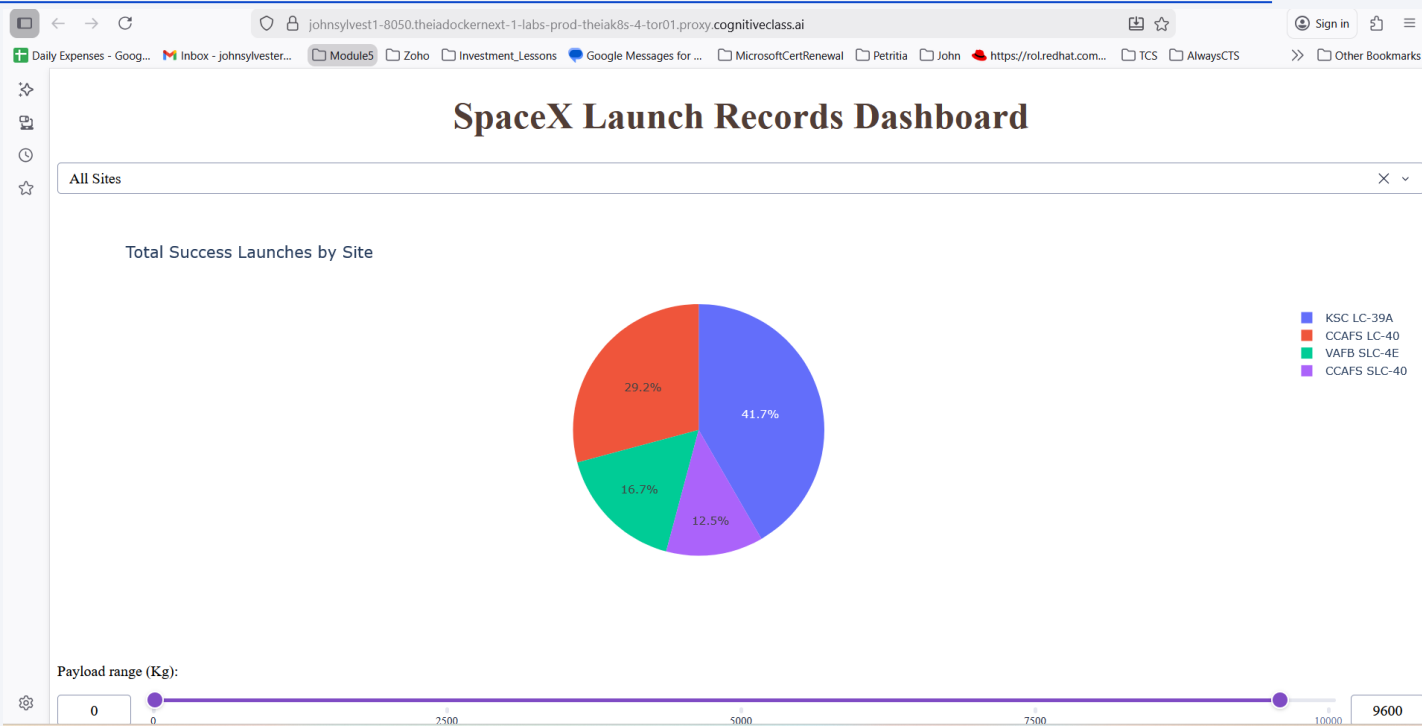
1. Primary Elements

- Site Filter (Dropdown):** Located at the top, currently set to "All Sites." This controls the data displayed in the pie chart below.
- Pie Chart ("Total Success Launches by Site"):** This is the central visual, representing the distribution of successful missions across four major launch facilities.
- Payload Range Slider:** At the bottom, there is a slider ranging from **0 to 10,000 kg**. It is currently set to its maximum range (**0 to 9600**), which filters the data based on the weight of the cargo carried by the rockets.

2. Key Findings from the Data

Based on the pie chart, we can see which sites are the most "productive" in terms of successful launches:

Launch Site	Success Contribution (%)	Performance Rank
KSC LC-39A (Kennedy Space Center)	41.7%	#1 (Highest)
CCAFS LC-40 (Cape Canaveral)	29.2%	#2
VAFB SLC-4E (Vandenberg AFB)	16.7%	#3
CCAFS SLC-40 (Secondary/Duplicate)	12.5%	#4

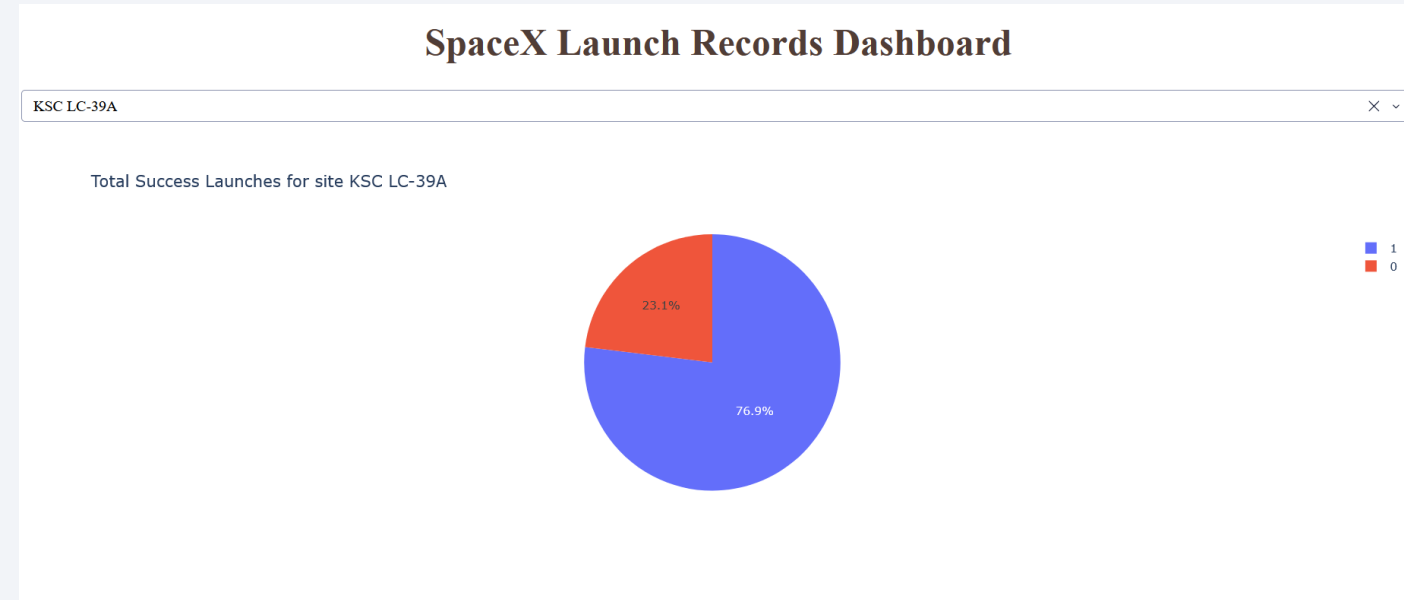


KSC LC -39A – Highest Launch success ratio

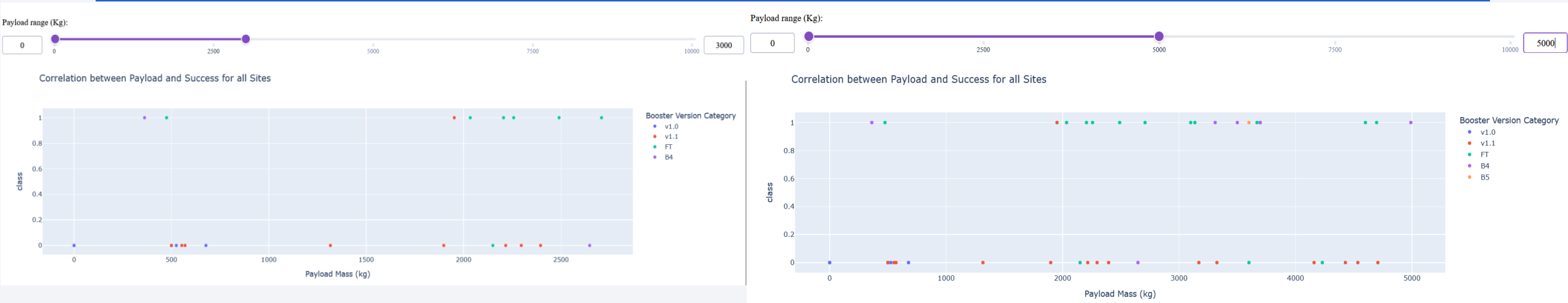
Key Findings for KSC LC-39A

The pie chart provides a clear performance metric for this specific launch complex:

- **Success Rate: 76.9%** of the recorded launches from this site were successful.
- **Failure Rate: 23.1%** of the recorded launches resulted in failure.
- **Overall Performance:** The site maintains a strong majority success rate. If we assume a standard dataset size often used in this specific data science project (typically 13 total launches for this site), this would translate to approximately **10 successes** and **3 failures**.



Payload vs. Launch Outcome scatter plot – All sites Slide 1

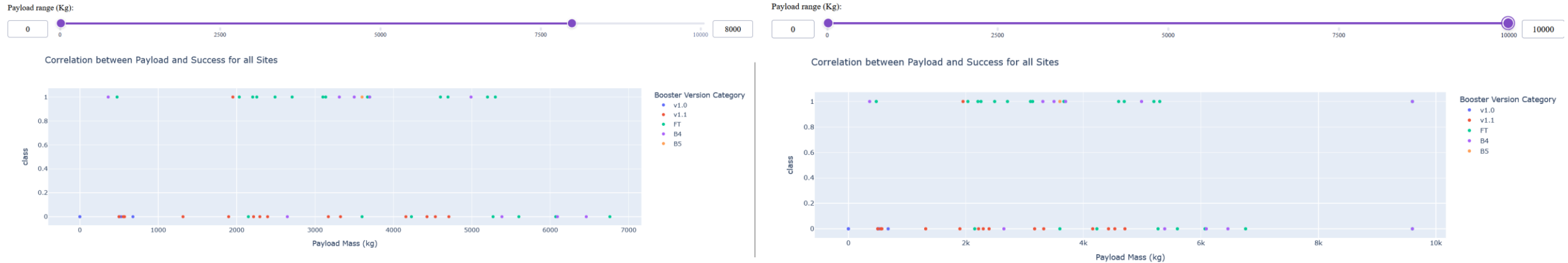


Key Findings by Payload Range

As we move the slider from **3,000 kg** to **10,000 kg**, several patterns emerge:

- **0 – 2,000 kg:** This range shows a high density of red (v1.1) and blue (v1.0) dots concentrated at **Class 0**. This indicates that early SpaceX missions, which typically carried lighter payloads, had a lower success rate compared to later iterations.
- **2,000 – 5,000 kg:** This is the **sweet spot for success** in this dataset. We see a significant cluster of green (FT) and purple (B4) dots at **Class 1**. This range historically represents many of the successful Falcon 9 Full Thrust missions.
- **5,000 – 7,000 kg:** As the payload increases, the number of data points thins out, but the success rate remains relatively high, particularly for the newer FT and B4 boosters.
- **8,000 – 10,000 kg:** There are very few launches in this ultra-heavy range within this specific dataset, but the few that exist (mostly B4 and B5 versions) show a strong leaning toward success.

Payload vs. Launch Outcome scatter plot – All sites Slide 2

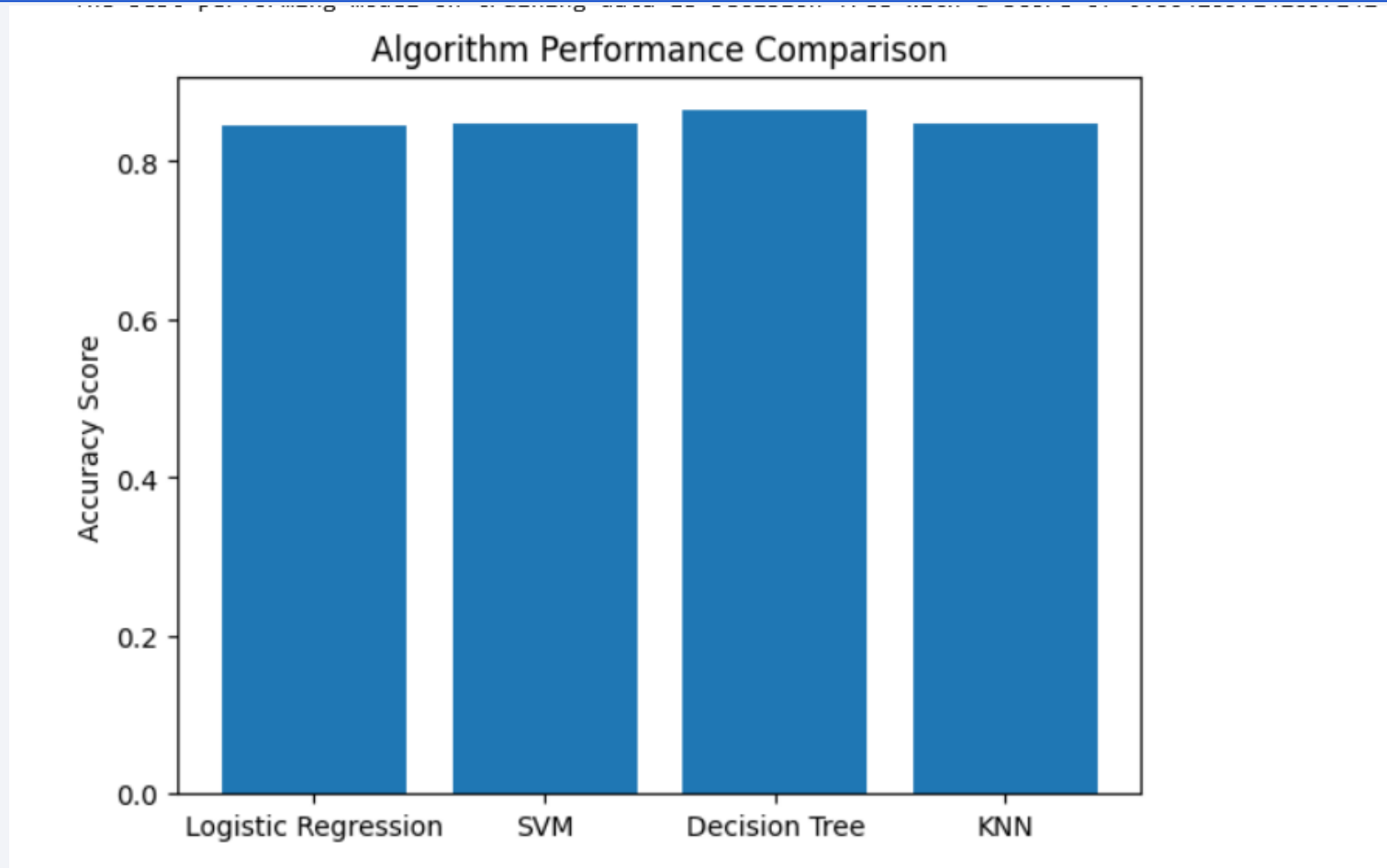


Booster Version	Category Color	Observation
v1.0	Blue	Mostly failures (Class 0) at very low payload masses.
v1.1	Red	Mixed results; many failures in the 500-2,500 kg range.
FT (Full Thrust)	Green	Highly Successful. This version dominates the Class 1 (Success) row between 2,000 kg and 6,000 kg.
B4 (Block 4)	Purple	Shows high reliability across various ranges, including the 10,000 kg mark.
B5 (Block 5)	Orange	The newest version shown; while fewer points are visible in this specific filtered view, it is historically the most successful and reliable version.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

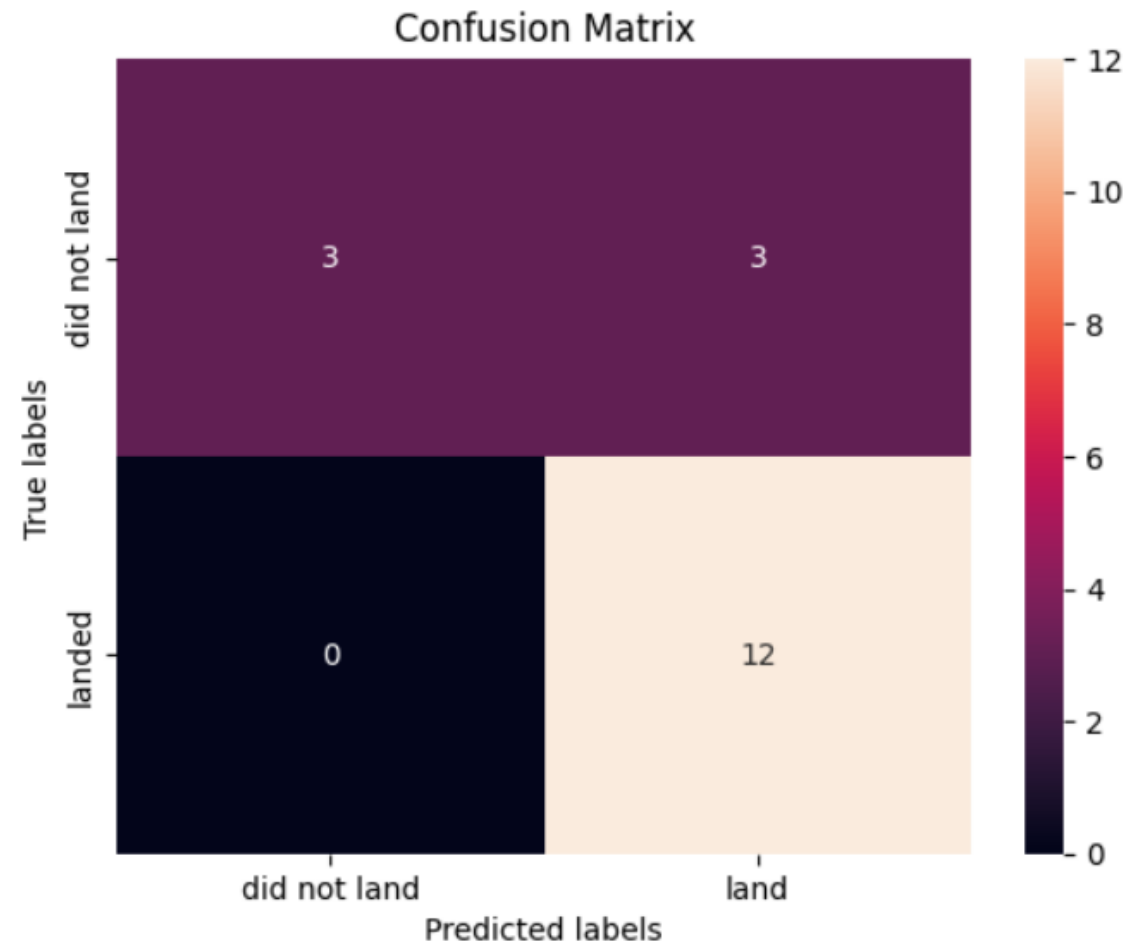


- The best performing model on training data is Decision Tree with a score of 0.8642857142857142

Confusion Matrix

Based on the confusion matrix results in the notebook, **Logistic Regression**, **SVM**, and **KNN** are the best-performing models, as they all achieved an identical test accuracy of **0.8333**.

- Here is an explanation of their performance:
- **High True Positive Rate:** These models demonstrated strong predictive power by correctly identifying 12 instances where the Falcon 9 first stage successfully landed.
- **Zero False Negatives:** The models were exceptionally reliable in their "did not land" predictions, as they never incorrectly predicted a failure when a successful landing actually occurred.
- **Identifiable Weakness:** The primary error across these top models was a small number of false positives, specifically 3 instances where they predicted a successful landing that ultimately failed.
- **Consistency through Tuning:** Through the use of GridSearchCV, these three distinct algorithms were optimized to reach the same peak performance level on the unseen test data.
- **Comparison to Decision Trees:** These models proved superior to the Decision Tree, which performed significantly worse with a test accuracy of only **0.6666**



Conclusions

- Identical Top Performance: Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) all achieved the highest and identical accuracy of approximately 83.33% on the test data.
- Predictive Accuracy on Success: All top-performing models (Logistic Regression, SVM, and KNN) successfully predicted 12 true positives, meaning they accurately identified 12 successful landings.
- Zero False Negatives: A key strength across the best models was the total absence of false negatives; they never predicted a landing would fail when it actually succeeded.
- Persistent False Positives: The primary error for the leading models was a set of 3 false positives, where the models incorrectly predicted a successful landing for instances that actually failed.
- Underperformance of Decision Trees: The Decision Tree model was the least effective algorithm, yielding a significantly lower test accuracy of 66.67% compared to the other three methods

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

